

P02 PROTECTED PHASE REVIEW

Analytics Engineering Platform

THE PROJECT LEARNING LOOP

- Open only the page matching the saved attempt.
- Compare evidence, not confidence or filenames.
- Diagnose the smallest weak competency.
- Close Review before a changed retry.
- Append the retry - never overwrite attempt-01.
- No numeric percentage shortcut.

PROTECTED REMEDIATION AID - NOT A SOLUTION BOOK

Build -> prove -> break -> repair -> rerun -> defend. No solved portfolio project is supplied.

Protected route rules

Before opening

A genuine phase attempt must already exist at the exact evidence path. This Review does not authorize creating an "attempt" from its wording.

Failure sequence

saved attempt -> targeted teaching repair -> changed micro-proof -> validation -> reassessment. The original attempt and first failure remain append-only.

Reviewer standard

Look for artifact reality, independent validation, changed/failure proof, runtime truth, explanation, and unresolved critical failure. A pretty README or green placeholder test is not enough.

Protected objective controls

Course QA also has objective controls in ``CONTROL/PROTECTED_CONTROLS/P02_expected_controls.json``. Use them only for reviewer diagnosis after attempt; they must not replace the learner's independent validation.

P02-01 - review after attempt

PASS evidence the reviewer should see

Define expected invariant before execution, run task, compare actual vs expected, and save one independent control check.

Artifact contract

business_questions.md + assumptions.md

Failure / transfer evidence

PHASE CHECK P02-01: one changed-input or broken-assumption test must be run and documented before mastery.

Targeted diagnosis

If weak, name only the failed competency. Likely causes for this phase: Starting with Airflow/dbt/Spark because they look impressive.; Writing vague goals such as "build a pipeline".; Hiding assumptions instead of recording them.. Use the recovery ladder: Rewrite the goal as: user + decision + source + output grain + freshness + failure consequence + evidence. If one field is unknown, mark it as an assumption rather than inventing certainty.

Fresh changed retry rule

Close this Review. Change one relevant input, assumption, failure condition, scale/security boundary, or source defect. Solve again without copying Review language. Re-run all previously correct critical controls and append retry evidence.

Reviewer questions

- Can I locate M18-P02-01-EVIDENCE and the real artifact?
- What independent control could expose a plausible-but-wrong result?
- What changed/bad input was tested?
- Which runtime claim was executed vs static/not executed?
- Can the learner explain why this design/control works and what would trigger redesign?

Move on only when

L3 independent professional; no unresolved critical failure; reproducible artifact/evidence.

P02-02 - review after attempt

PASS evidence the reviewer should see

Record row counts, grain, keys, null/duplicate/schema defects; independently sample-check at least one finding.

Artifact contract

source_profile/ + grain_key_defects.md

Failure / transfer evidence

Replay the same batch/event plus one late/changed record; detect duplicate/logical-state corruption if implementation is unsafe.

Targeted diagnosis

If weak, name only the failed competency. Likely causes for this phase: Calling every repeated customer_id a duplicate.; Trusting automated counts without a manual spot check.; Skipping timestamp/timezone or type inspection.. Use the recovery ladder: State the grain first. Then list candidate keys, count rows/distinct keys, inspect null/type/schema issues, and manually verify one defect before moving on.

Fresh changed retry rule

Close this Review. Change one relevant input, assumption, failure condition, scale/security boundary, or source defect. Solve again without copying Review language. Re-run all previously correct critical controls and append retry evidence.

Reviewer questions

- Can I locate M18-P02-02-EVIDENCE and the real artifact?
- What independent control could expose a plausible-but-wrong result?
- What changed/bad input was tested?
- Which runtime claim was executed vs static/not executed?
- Can the learner explain why this design/control works and what would trigger redesign?

Move on only when

L3 independent professional; no unresolved critical failure; reproducible artifact/evidence.

P02-03 - review after attempt

PASS evidence the reviewer should see

Define expected invariant before execution, run task, compare actual vs expected, and save one independent control check.

Artifact contract

p02_phase_artifact + evidence note

Failure / transfer evidence

PHASE CHECK P02-03: one changed-input or broken-assumption test must be run and documented before mastery.

Targeted diagnosis

If weak, name only the failed competency. Likely causes for this phase: Drawing products without responsibilities.; No replay path from raw evidence.; No rejected alternative or revisit trigger.. Use the recovery ladder: Trace one source from arrival to serving. At each arrow ask: what is persisted, what can fail, what validates it, and how would I replay/recover?

Fresh changed retry rule

Close this Review. Change one relevant input, assumption, failure condition, scale/security boundary, or source defect. Solve again without copying Review language. Re-run all previously correct critical controls and append retry evidence.

Reviewer questions

- Can I locate M18-P02-03-EVIDENCE and the real artifact?
- What independent control could expose a plausible-but-wrong result?
- What changed/bad input was tested?
- Which runtime claim was executed vs static/not executed?
- Can the learner explain why this design/control works and what would trigger redesign?

Move on only when

L3 independent professional; no unresolved critical failure; reproducible artifact/evidence.

P02-04 - review after attempt

PASS evidence the reviewer should see

State output grain; test key/cardinality; reconcile critical measures to an independent control query.

Artifact contract

models/SQL/dbt + lineage + row-count controls

Failure / transfer evidence

PHASE CHECK P02-04: one changed-input or broken-assumption test must be run and documented before mastery.

Targeted diagnosis

If weak, name only the failed competency. Likely causes for this phase: Aggregating too early in staging.; Using DISTINCT to hide fan-out.; Building marts without documenting lineage.. Use the recovery ladder: Run each model/stage alone. State one-row meaning. Check keys and row counts before the join, after the join, and against an independent control.

Fresh changed retry rule

Close this Review. Change one relevant input, assumption, failure condition, scale/security boundary, or source defect. Solve again without copying Review language. Re-run all previously correct critical controls and append retry evidence.

Reviewer questions

- Can I locate M18-P02-04-EVIDENCE and the real artifact?
- What independent control could expose a plausible-but-wrong result?
- What changed/bad input was tested?
- Which runtime claim was executed vs static/not executed?
- Can the learner explain why this design/control works and what would trigger redesign?

Move on only when

L3 independent professional; no unresolved critical failure; reproducible artifact/evidence.

P02-05 - review after attempt

PASS evidence the reviewer should see

State output grain; test key/cardinality; reconcile critical measures to an independent control query.

Artifact contract

source_contracts.md

Failure / transfer evidence

Inject one duplicate dimension/reference key or one-to-many fan-out risk and prove the validation catches it.

Targeted diagnosis

If weak, name only the failed competency. Likely causes for this phase: Aggregating too early in staging.; Using DISTINCT to hide fan-out.; Building marts without documenting lineage.. Use the recovery ladder: Run each model/stage alone. State one-row meaning. Check keys and row counts before the join, after the join, and against an independent control.

Fresh changed retry rule

Close this Review. Change one relevant input, assumption, failure condition, scale/security boundary, or source defect. Solve again without copying Review language. Re-run all previously correct critical controls and append retry evidence.

Reviewer questions

- Can I locate M18-P02-05-EVIDENCE and the real artifact?
- What independent control could expose a plausible-but-wrong result?
- What changed/bad input was tested?
- Which runtime claim was executed vs static/not executed?
- Can the learner explain why this design/control works and what would trigger redesign?

Move on only when

L3 independent professional; no unresolved critical failure; reproducible artifact/evidence.

P02-06 - review after attempt

PASS evidence the reviewer should see

State output grain; test key/cardinality; reconcile critical measures to an independent control query.

Artifact contract

models/SQL/dbt + lineage + row-count controls

Failure / transfer evidence

Inject one duplicate dimension/reference key or one-to-many fan-out risk and prove the validation catches it.

Targeted diagnosis

If weak, name only the failed competency. Likely causes for this phase: Aggregating too early in staging.; Using DISTINCT to hide fan-out.; Building marts without documenting lineage.. Use the recovery ladder: Run each model/stage alone. State one-row meaning. Check keys and row counts before the join, after the join, and against an independent control.

Fresh changed retry rule

Close this Review. Change one relevant input, assumption, failure condition, scale/security boundary, or source defect. Solve again without copying Review language. Re-run all previously correct critical controls and append retry evidence.

Reviewer questions

- Can I locate M18-P02-06-EVIDENCE and the real artifact?
- What independent control could expose a plausible-but-wrong result?
- What changed/bad input was tested?
- Which runtime claim was executed vs static/not executed?
- Can the learner explain why this design/control works and what would trigger redesign?

Move on only when

L3 independent professional; no unresolved critical failure; reproducible artifact/evidence.

P02-07 - review after attempt

PASS evidence the reviewer should see

Define expected invariant before execution, run task, compare actual vs expected, and save one independent control check.

Artifact contract

tests/ + reconciliation report + quarantine evidence

Failure / transfer evidence

PHASE CHECK P02-07: one changed-input or broken-assumption test must be run and documented before mastery.

Targeted diagnosis

If weak, name only the failed competency. Likely causes for this phase: Only testing happy-path row counts.; Using DISTINCT to hide duplication.; Letting tests exist without a publish consequence.. Use the recovery ladder: Write the invariant in plain English, define failure action, run it on a known good and known bad example, then reconcile independently.

Fresh changed retry rule

Close this Review. Change one relevant input, assumption, failure condition, scale/security boundary, or source defect. Solve again without copying Review language. Re-run all previously correct critical controls and append retry evidence.

Reviewer questions

- Can I locate M18-P02-07-EVIDENCE and the real artifact?
- What independent control could expose a plausible-but-wrong result?
- What changed/bad input was tested?
- Which runtime claim was executed vs static/not executed?
- Can the learner explain why this design/control works and what would trigger redesign?

Move on only when

L3 independent professional; no unresolved critical failure; reproducible artifact/evidence.

P02-08 - review after attempt

PASS evidence the reviewer should see

State output grain; test key/cardinality; reconcile critical measures to an independent control query.

Artifact contract

state ledger/watermark + rerun/backfill evidence

Failure / transfer evidence

Replay the same batch/event plus one late/changed record; detect duplicate/logical-state corruption if implementation is unsafe.

Targeted diagnosis

If weak, name only the failed competency. Likely causes for this phase: Using file name alone as business identity.; Advancing watermark before downstream success.; Treating late data as duplicate by default.. Use the recovery ladder: Write identity key, ordering rule, state update moment, and replay behavior before code. Test same-input rerun, late input, and bounded backfill separately.

Fresh changed retry rule

Close this Review. Change one relevant input, assumption, failure condition, scale/security boundary, or source defect. Solve again without copying Review language. Re-run all previously correct critical controls and append retry evidence.

Reviewer questions

- Can I locate M18-P02-08-EVIDENCE and the real artifact?
- What independent control could expose a plausible-but-wrong result?
- What changed/bad input was tested?
- Which runtime claim was executed vs static/not executed?
- Can the learner explain why this design/control works and what would trigger redesign?

Move on only when

L3 independent professional; no unresolved critical failure; reproducible artifact/evidence.

P02-09 - review after attempt

PASS evidence the reviewer should see

Define expected invariant before execution, run task, compare actual vs expected, and save one independent control check.

Artifact contract

p02_phase_artifact + evidence note

Failure / transfer evidence

PHASE CHECK P02-09: one changed-input or broken-assumption test must be run and documented before mastery.

Targeted diagnosis

If weak, name only the failed competency. Likely causes for this phase: Using file name alone as business identity.; Advancing watermark before downstream success.; Treating late data as duplicate by default.. Use the recovery ladder: Write identity key, ordering rule, state update moment, and replay behavior before code. Test same-input rerun, late input, and bounded backfill separately.

Fresh changed retry rule

Close this Review. Change one relevant input, assumption, failure condition, scale/security boundary, or source defect. Solve again without copying Review language. Re-run all previously correct critical controls and append retry evidence.

Reviewer questions

- Can I locate M18-P02-09-EVIDENCE and the real artifact?
- What independent control could expose a plausible-but-wrong result?
- What changed/bad input was tested?
- Which runtime claim was executed vs static/not executed?
- Can the learner explain why this design/control works and what would trigger redesign?

Move on only when

L3 independent professional; no unresolved critical failure; reproducible artifact/evidence.

P02-10 - review after attempt

PASS evidence the reviewer should see

Run dependency path end-to-end; prove failed task does not falsely advance downstream state; rerun only required work.

Artifact contract

DAG/workflow + run log + dependency evidence

Failure / transfer evidence

Force an upstream task failure/partial run; prove downstream publish does not falsely succeed and rerun is bounded.

Targeted diagnosis

If weak, name only the failed competency. Likely causes for this phase: Retries around non-idempotent side effects.; Downstream tasks running after partial upstream success.; Backfilling all history when only one window is damaged.. Use the recovery ladder: For each task write input, output, success condition, retry safety, and downstream dependency. Then inject one upstream failure before trusting automation.

Fresh changed retry rule

Close this Review. Change one relevant input, assumption, failure condition, scale/security boundary, or source defect. Solve again without copying Review language. Re-run all previously correct critical controls and append retry evidence.

Reviewer questions

- Can I locate M18-P02-10-EVIDENCE and the real artifact?
- What independent control could expose a plausible-but-wrong result?
- What changed/bad input was tested?
- Which runtime claim was executed vs static/not executed?
- Can the learner explain why this design/control works and what would trigger redesign?

Move on only when

L3 independent professional; no unresolved critical failure; reproducible artifact/evidence.

P02-11 - review after attempt

PASS evidence the reviewer should see

Save first failing evidence before repair; identify root cause; apply smallest safe repair; fresh retry must pass all prior controls.

Artifact contract

first-failure evidence + root_cause.md + repair + fresh retry evidence

Failure / transfer evidence

MANDATORY SUPPLIED FAILURE: capture first failure, diagnose, repair, fresh retry; do not overwrite original evidence.

Targeted diagnosis

If weak, name only the failed competency. Likely causes for this phase: Deleting the failed run.; Changing multiple things at once.; Calling symptom removal a root cause.. Use the recovery ladder: Freeze failure evidence -> state expected vs actual -> isolate boundary -> test one hypothesis -> smallest repair -> changed retry -> re-run previous critical controls.

Fresh changed retry rule

Close this Review. Change one relevant input, assumption, failure condition, scale/security boundary, or source defect. Solve again without copying Review language. Re-run all previously correct critical controls and append retry evidence.

Reviewer questions

- Can I locate M18-P02-11-EVIDENCE and the real artifact?
- What independent control could expose a plausible-but-wrong result?
- What changed/bad input was tested?
- Which runtime claim was executed vs static/not executed?
- Can the learner explain why this design/control works and what would trigger redesign?

Move on only when

L3 independent professional; no unresolved critical failure; reproducible artifact/evidence.

P02-12 - review after attempt

PASS evidence the reviewer should see

Reviewer can locate every evidence artifact and challenge one design trade-off; learner explains limitation and recovery path without reading a script.

Artifact contract

README.md + runbook.md + evidence_index.md

Failure / transfer evidence

PHASE CHECK P02-12: one changed-input or broken-assumption test must be run and documented before mastery.

Targeted diagnosis

If weak, name only the failed competency. Likely causes for this phase: CI that only checks syntax.; Monitoring without response threshold/action.; Rollback plan that cannot restore data/state safely.. Use the recovery ladder: Trace change -> test -> approve -> deploy -> observe -> rollback. If any arrow has no evidence or owner, production readiness is incomplete.

Fresh changed retry rule

Close this Review. Change one relevant input, assumption, failure condition, scale/security boundary, or source defect. Solve again without copying Review language. Re-run all previously correct critical controls and append retry evidence.

Reviewer questions

- Can I locate M18-P02-12-EVIDENCE and the real artifact?
- What independent control could expose a plausible-but-wrong result?
- What changed/bad input was tested?
- Which runtime claim was executed vs static/not executed?
- Can the learner explain why this design/control works and what would trigger redesign?

Move on only when

L3 independent professional; no unresolved critical failure; reproducible artifact/evidence.

P02-13 - review after attempt

PASS evidence the reviewer should see

Rebuild from documented clean state using only declared dependencies/config; reviewer can reproduce declared outputs.

Artifact contract

clean-clone transcript/log + frozen evidence bundle

Failure / transfer evidence

PHASE CHECK P02-13: one changed-input or broken-assumption test must be run and documented before mastery.

Targeted diagnosis

If weak, name only the failed competency. Likely causes for this phase: Demo works only on the original machine state.; Reading README verbatim as defense.; Claiming unexecuted cloud/runtime work.. Use the recovery ladder: Use defense order: business goal -> source/grain -> architecture -> critical controls -> failure/recovery -> runtime truth -> biggest trade-off -> revisit trigger.

Fresh changed retry rule

Close this Review. Change one relevant input, assumption, failure condition, scale/security boundary, or source defect. Solve again without copying Review language. Re-run all previously correct critical controls and append retry evidence.

Reviewer questions

- Can I locate M18-P02-13-EVIDENCE and the real artifact?
- What independent control could expose a plausible-but-wrong result?
- What changed/bad input was tested?
- Which runtime claim was executed vs static/not executed?
- Can the learner explain why this design/control works and what would trigger redesign?

Move on only when

L3 independent professional; no unresolved critical failure; reproducible artifact/evidence.